Ravichandran P

Computer Science and Engineering Student

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Objective

Computer Science undergraduate specializing in AI/ML and full-stack development, with hands-on experience building end-to-end systems (satellite tracking, speech-authenticity detection) and leading award-shortlisted teams at national hackathons. Seeking a Software Engineer role to apply strong problem-solving skills and a track record of shipping working products under deadline pressure.

Technical Skills

Languages: Java, Python
Web Technologies: HTML, CSS, Flask, FastAPI
AI / ML: Machine Learning, Deep Learning, NLP, Generative AI, TensorFlow, OpenCV
Databases: SQL
Tools & Platforms: Git/GitHub, VS Code, N8N, MySQL.

Education

Bachelor of Engineering – Computer Science and Engineering 2023 – 2027
Kongunadu College of Engineering and Technology, Namakkal 75%

HSC 2022 – 2023
Kalaimagal Matriculation Higher Secondary School 78%

SSLC 2020 – 2021
Kalaimagal Matriculation Higher Secondary School 100%

Projects

LEO Satellite Visibility & GPS-Denied Navigation System

- Built a full-stack LEO satellite visibility and TLE filtering system using **Python, Flask, Skyfield, SGP4, JavaScript, HTML/CSS** and real-time satellite orbital data.
- Implemented TLE-based orbit propagation and real-time visibility prediction using observer location, elevation angle, and satellite orbital parameters.
- Developed a **GPS-denied navigation approach** using LEO satellite signals and Doppler-based measurements as an alternative positioning source when GNSS/GPS is unavailable.
- Built an interactive web dashboard to visualize visible satellites, orbital data, elevation, and predicted passes for end users.

AI-Generated Voice Detection System

- Built an AI speech-authenticity detection system using **Python, Flask, TensorFlow, NumPy, and REST APIs**, classifying AI-generated vs. human speech with confidence scores.
- Extracted **MFCCs, Mel-spectrograms, pitch, and energy features** from audio signals to improve classification performance.
- Designed a secure REST API supporting Base64-encoded audio input and structured JSON responses for external integration.
- Implemented a **CNN-based audio classification pipeline** to analyze spectrograms and identify patterns associated with synthetic speech.

AI Developer — AI Agent Automation System

- Developed autonomous **AI agents** capable of executing multi-step workflows using **n8n automation** and LLM-based reasoning.
- Integrated **LLM-driven decision-making** with API-based actions to automate tasks, information retrieval, and workflow execution.

- Implemented **webhooks, REST APIs, and Ngrok** for communication between AI agents and external services.
- Designed scalable workflow architectures for **task automation, information processing, and decision support**.

IDEA Lab Website

- Designed and developed the college's official IDEA Lab website to showcase innovation projects, events, and lab activities.
- Built responsive pages ensuring compatibility across desktop and mobile.
- Improved UX through intuitive navigation and optimized page performance.

Achievements

HCL GUVI AI Impact Summit—Buildthon Finalist 2026

- Led a team in developing an AI-powered solution from ideation to implementation and final presentation.
- Coordinated technical tasks, team responsibilities, and solution refinement under competition deadlines.

Smart India Hackathon (SIH) – National Finalist 2025

- Contributed to a national-level finalist project, collaborating on solution development and technical implementation.
- Worked with the team to analyze the problem statement, develop the proposed solution, and prepare the final presentation.

Cognizant Technoverse Hackathon 2026

- Led a team through multiple stages of a national-level hackathon, advancing to the second round.
- Managed technical execution, task allocation, and collaborative problem-solving throughout the competition.

CMR Hackathon – 2nd Runner-Up 2026

- Recognized among 1,000+ participating teams for technical innovation and effective solution development.
- Secured **2nd Runner-Up** position and received a **Rs.,60,000 prize** for the project.
- Led the team in developing, refining, and presenting the solution to the judging panel.

Internship

Python & ML Intern – Inspire Softech Solutions Nov – Dec 2024

- Gained hands-on training in **Python programming and Machine Learning** through practical exercises and guided development tasks.
- Built a mini project applying core ML concepts, including data preprocessing, feature extraction, model training, and evaluation.
- Worked with Python-based libraries and tools to implement and test machine learning workflows on real-world datasets.
- Strengthened practical understanding of **data analysis, model development, debugging, and problem-solving** through hands-on training.

Certifications

Frontend Development Essentials – CodeKaro
Generative AI & GPT Models – Infosys

Leadership & Involvement

- Joint Secretary, CSE Department – supported event planning, student coordination, and departmental activities.
- Organized CELESTA 2K25, an inter-college technical symposium, coordinating competitions and hackathons.
- Led teams in multiple national-level hackathons, managing collaboration, task allocation, and delivery.
- Contributed to departmental R&D activities, assisting in technical research and project implementation.

Area of Interest

Full-Stack Development, Software Engineering, Artificial Intelligence

Hobbies

Tech content creation, video editing, and Discord community engagement.